MCQ solutions

1. **B.** $u_1 = 5$ then $u_2 = 8$. Answers 5 and 11 confuse the indices; $2^4 = 16$ does not correspond to the additive relation.
2. **C.** We fix an index n and we assume $P(n)$ to prove $P(n+1)$. Checking the first index is the base case, and directly assuming the conclusion would be circular.
3. **B.** The indices are 0, 1, 2, 3, 4: five terms. The number of terms is last index minus first index plus one.
4. **B.** The sign of the differences proves the sequence is strictly increasing. $u_n = n - 100$ shows that neither positivity nor an upper bound nor convergence is implied.
5. **C.** The exit occurs when the condition $u \leq 10$ becomes false, so for $u > 10$. Reaching exactly 10 is not enough to stop the loop.
6. **B.** Monotone convergence gives a limit at most equal to 7. The constant sequence -1 refutes the limit 7 and positivity; $7 - 1/(n+1)$ is not geometric.
7. **A.** A value left unchanged by the update satisfies $L = 0{,}5L + 4$, i.e. $L = 8$. This equation alone does not prove convergence.
8. **B.** $v_{n+1} = 0{,}5u_n + 4 - 8 = 0{,}5(u_n - 8) = 0{,}5v_n$. Subtracting the fixed point removes the constant term.
9. **B.** The gap $|u_n - 6|$ measures the distance from u_n to 6. The strict inequality $|u_n - 6| < 0{,}01$ describes exactly the required open interval.
10. **C.** A finite calculation tells us only about the terms actually computed. A property for every index requires a proof; convergence requires a study of the whole sequence.

True/false. False: $u_n = n$ diverges. True: $2^{-n} > 0$, hence $6 - 2^{-n} < 6$. False: with $f(x) = -x$, the fixed point is 0 but $u_0 = 1$ generates $(-1)^n$, which diverges.

Solutions to the applications and guidance

Application 1. $v_0 = 1$, $v_1 = 3$, $v_2 = 7$, $v_3 = 15$. The points are $(0;1), (1;3), (2;7), (3;15)$. $v_2 + 1 = 8$ but $v_{2+1} = 15$.

Application 2. Three updates: repeat `u=3*u-1`, then total=total+u. We get $2 + 5 + 14 + 41 = 62$. The reverse order would successively add 2,5,14 to the total already initialised to 2 and would omit 41.

Application 3. At index 0, $3 \times 2^0 - 1 = 2 = a_0$. Fix $n \in \mathbb{N}$ and suppose $a_n = 3 \times 2^n - 1$. Then $a_{n+1} = 2(3 \times 2^n - 1) + 1 = 3 \times 2^{n+1} - 1$. The base case and the inductive step allow us to conclude for every $n \in \mathbb{N}$.

Application 4. $u_0 = 0$ gives the zero sequence: the inductive step is correct but never starts. Add $u_0 \geq 2$, then check $P(0)$. Do not confuse the validity of an implication with that of its hypothesis.

Application 5. At index 0: $1 = 1^2$. Under the hypothesis at index n, the sum at index $n + 1$ equals $(n+1)^2 + 2(n+1) + 1 = (n+2)^2$. Conclude by induction. The new term has index $n + 1$.

Application 6. Arithmetic sequence $a_n = 40 + 5n$; geometric sequence $b_n = 40(1{,}1)^n$. First costs : 40,45,50 and 40,44,48.4. Cumulative totals: $10(40 + 85)/2 = 625$ euros; 40 $\frac{1{,}1^{10}-1}{0{,}1} \simeq 637{,}50$ euros. The three first years are not enough to predict the order of the cumulative costs.

Class discussion and differentiation

Have students compare their working in pairs before showing the formula. For the gap-fill proof, accept a personal wording provided the integer is fixed and the hypothesis is identifiable.
Then remove the prompts. At S13, go back to the base-case error with $u_0 = -1$. Observe the write-up and not only the numerical result.

Teaching notes and solutions

Guidance for the teacher

In S2, distinguish index and value of a term from the start. In S4, have students run a loop by hand before Python. In S5 and S7, have students say aloud "assume at index n, we prove at index $n+1$". Sums and divisibility come in S10, after DS1: do not assess them in the first written test. In S11, finding the fixed point is an algebraic technique; the theorem of convergence will be established in the next chapter. For struggling students, provide the property $P(n)$ and let them write out the three steps; for the more advanced, remove the hint about the auxiliary sequence.

Solutions to the activity

1. $u_0 = 100$ L. After one day, there are $0{,}8u_0$ L left, then 30 L are added: $u_1 = 110$ L. Similarly, $u_2 = 0{,}8 \times 110 + 30 = 118$ L. For every $n \geq 0$, $u_{n+1} = 0{,}8u_n + 30$.

2. The equilibrium volume L should satisfy $L = 0{,}8L + 30$, i.e. $L = 150$ L. Let us prove $P(n)$: "$100 \leq u_n < 150$". At index 0, $u_0 = 100 \in [100, 150[$. Fix $n \geq 0$ and suppose $P(n)$ true. Since $0{,}8 > 0$,
$$110 = 0{,}8 \times 100 + 30 \leq 0{,}8u_n + 30 < 0{,}8 \times 150 + 30 = 150.$$
Hence $100 \leq u_{n+1} < 150$ and $P(n+1)$ is true. By induction, the bounds hold for all $n \in \mathbb{N}$.

3. $v_{n+1} = u_{n+1} - 150 = 0{,}8u_n - 120 = 0{,}8(u_n - 150) = 0{,}8v_n$. The sequence (v_n) is geometric, with common ratio 0.8 and first term $v_0 = -50$. Thus $v_n = -50(0{,}8)^n$, then $u_n = 150 - 50(0{,}8)^n$.

4.
```
def premier_jour():
    n, u = 0, 100
    while u <= 145:
        u = 0.8*u+30
        n += 1
    return n
```
The loop returns 11. Indeed, $u_{10} \simeq 144{,}63 < 145$ and $u_{11} \simeq 145{,}71 > 145$. Moreover, $u_{n+1} - u_n = 10(0{,}8)^n > 0$: the threshold was not crossed earlier. Computing these two values is enough here to guarantee the loop stops, without yet requiring a convergence theorem.

5. The simulation only computes a finite number of terms. The inductive step, together with the base case, establishes the bounds for all integers; it is this that rules out exceeding 150 L.

E1. $u_4 = 5 - 3 \times 4 = -7$. The sequence is arithmetic with common difference -3 and $u_{20} = -55$. The sum has 21 terms: $\sum_{k=0}^{20} u_k = \frac{21(u_0+u_{20})}{2} = \frac{21(5-55)}{2} = -525$.

E2. The sequence is geometric with common ratio 2 and first term $v_0 = 3$. Hence $v_n = 3 \times 2^n$. The sum contains $n+1$ terms and equals $\frac{3(1-2^{n+1})}{1-2} = 3(2^{n+1} - 1)$.

E3. Let $P(n)$: "$3^n \geq 2n+1$". At index 0, $3^0 = 1 = 2 \times 0 + 1$. Fix $n \geq 0$ and suppose $P(n)$ true. Multiplying by $3 > 0$, $3^{n+1} \geq 6n + 3$. Now $6n + 3 - (2n+3) = 4n \geq 0$. Thus $3^{n+1} \geq 2(n+1) + 1$, which is $P(n+1)$. By induction, $P(n)$ is true for all $n \in \mathbb{N}$.

E4. For $n \geq 0$, $u_n = 1 - \frac{1}{n+2}$. We have $0 < \frac{1}{n+2} \leq \frac{1}{2}$, therefore $\frac{1}{2} \leq u_n < 1$. Moreover,
$$u_{n+1} - u_n = \frac{1}{n+2} - \frac{1}{n+3} = \frac{1}{(n+2)(n+3)} > 0.$$
The sequence is strictly increasing, bounded below by $\frac{1}{2}$ and bounded above by 1.

E5. Let $P(n)$: "there exists an integer k such that $8^n - 1 = 7k$ ". For $n = 0$, $8^0 - 1 = 0 = 7 \times 0 : P(0)$ is true. Fix $n \geq 0$ and suppose $P(n)$ true. So there exists an integer k such that $8^n - 1 = 7k$. Then

$$8^{n+1} - 1 = 8(8^n - 1) + 7 = 8 \times 7k + 7 = 7(8k + 1).$$

Since $8k + 1$ is an integer, $P(n+1)$ is true. By induction, 7 divides $8^n - 1$ for every $n \geq 0$.

Solutions to homework DM1

1. Let $P(n)$: "$2 \leq u_n < 6$". At index 0, $u_0 = 2$ satisfies the bounds. Fix $n \geq 0$ and suppose $P(n)$ true. Then $4 \leq \frac{u_n}{2} + 3 < 6$; in particular $2 \leq u_{n+1} < 6$. By induction, $P(n)$ is true at every index. We deduce $u_{n+1} - u_n = 3 - \frac{u_n}{2} = \frac{6-u_n}{2} > 0$. The sequence is strictly increasing.

2. The equation $L = \frac{L}{2} + 3$ gives $L = 6$. Let $v_n = u_n - 6$. Then $v_{n+1} = \frac{u_n}{2} - 3 = \frac{v_n}{2}$ & $v_0 = -4$. Hence $v_n = -\frac{4}{2^n}$ and $u_n = 6 - \frac{4}{2^n}$.

3. By linearity of sums and the geometric sum formula,

$$\begin{aligned} S_n &= 6(n+1) - 4\sum_{k=0}^{n}(\frac{1}{2})^k \\ &= 6(n+1) - 8(1 - 2^{-(n+1)}). \end{aligned}$$

4. $u_n > 5{,}99$ is equivalent to $\frac{4}{2^n} < 0{,}01$, i.e. $2^n > 400$. Since $2^8 = 256 \leq 400 < 512 = 2^9$ and (2^n) is increasing, the smallest index is 9.

5. The mean is less than 6 and its distance from 6 is

$$6 - \frac{S_n}{n+1} = \frac{8(1 - 2^{-(n+1)})}{n+1}, \qquad 0 < 6 - \frac{S_n}{n+1} < \frac{8}{n+1}.$$

To make this difference less than a precision $\varepsilon > 0$, it suffices to take $n + 1 > \frac{8}{\varepsilon}$. This is a quantitative control of the gap, which does not yet require a theorem on limits.

Solutions to the textbook or past-paper work

Exercise 22. The four translations are, for all $n \in \mathbb{N}$: $u_{n+1} \geq u_n$; $v_n \geq 5$; $w_n \leq -1$; $t_n > 0$. The monotonicity compares two consecutive terms; a bound compares each term with the same number.

Exercise 25. We have $u_0 = -5$ and $u_{n+1} = \frac{2}{3}u_n - 1$.

Base case. $u_0 = -5 \leq -3$: the property is true at index 0.

Inductive step. Fix $n \in \mathbb{N}$ and suppose $u_n \leq -3$. Multiplication by $\frac{2}{3} > 0$ preserves order, so $\frac{2}{3}u_n \leq -2$. Thus $u_{n+1} = \frac{2}{3}u_n - 1 \leq -3$.

Conclusion. By induction, $u_n \leq -3$ for every $n \in \mathbb{N}$. Then, $u_{n+1} - u_n = \frac{2}{3}u_n - 1 - u_n = -\frac{1}{3}u_n - 1$. The previous bound gives $-\frac{1}{3}u_n \geq 1$, hence $u_{n+1} - u_n \geq 0$. The sequence is increasing. The change of direction of the inequality here comes from the negative factor $-\frac{1}{3}$.